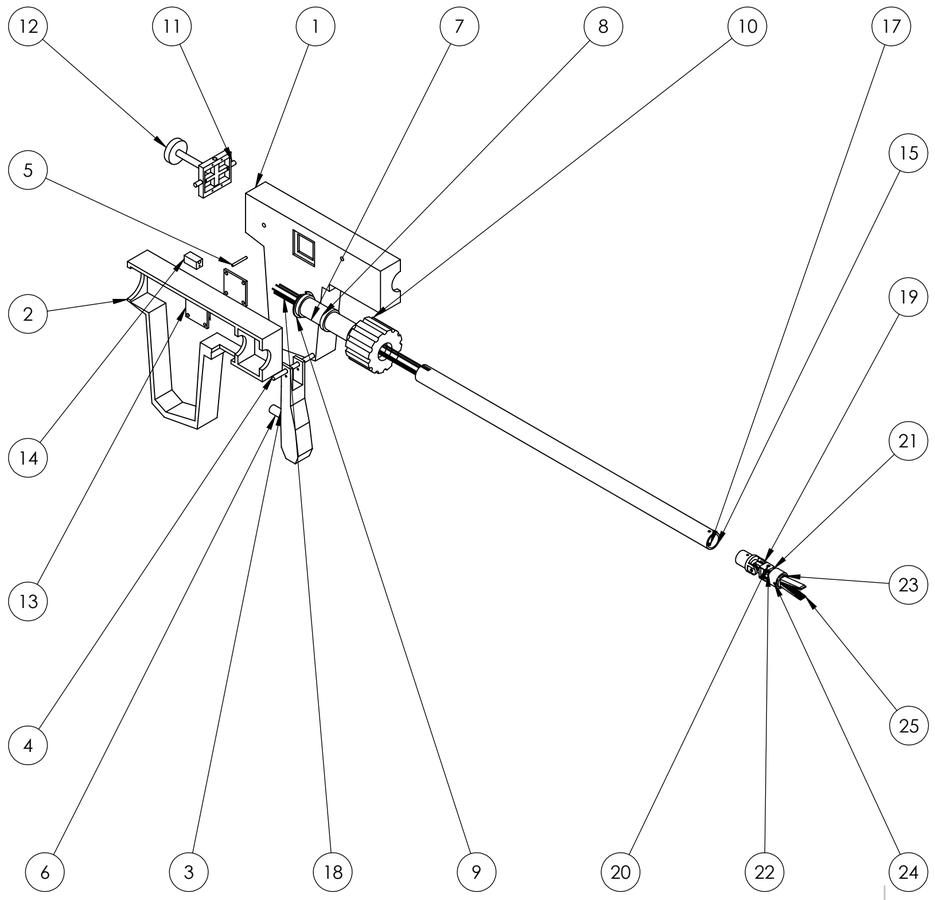




ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	HA-001_Housing_Left	Handle housing, left half, snap/screw boss shell	1
2	HA-002_Housing_Right	Handle housing, right half (mirror of HA-001)	1
3	HA-003_Trigger	Jaw trigger lever, 0 to -12 deg, cable anchor arm	1
4	HA-004_Trigger_Pivot_Pin	Trigger pivot pin D3 h8, retained by housings	1
5	HA-014_Cable_Anchor_Pin	Cable anchor cross-pin for trigger tendon	1
6	HA-005_Trigger_Return_Spring	Trigger return compression spring (simplified solid)	1
7	HA-007_Rotation_Hub	Rotation hub, shaft bonded, runs in PEEK bushings	1
8	HA-008_PEEK_Bushing	Flanged plain bushing D9 H8/f7 for rotation hub	2
9	HA-009_E_Clip	E-clip axial retainer for rotation hub (simplified)	1
10	HA-006_Rotation_Knob	Shaft rotation knob, knurled collar	1
11	HA-011_Gimbal_Yoke	2-axis gimbal yoke with pitch/yaw cable drums R2.2	1
12	HA-010_Wrist_Lever	Wrist control lever (thumb joystick stem)	1
13	HA-012_Cover_Plate	Housing access cover plate (qty 2)	2
14	HA-013_Cable_Tensioner	Threaded cable tensioner / pretension block	1
15	SH-001_Shaft_Tube_RevA	Instrument shaft tube 8.5 OD x 0.8 wall x 190	1
16	SH-002_Cable_Guide_Spacer_RevB	Multi-lumen cable guide spacer, D6.85 x 8, bonded in shaft (qty 3)	3
17	SH-003_Wrist_Retaining_Pin	Wrist base retaining cross-pin D1.0 x 8.4, laser-welded flush	2
18	CB-001_Tendon_D045	Tendon 7x19 SS wire rope D0.45 (representative straight run, cut to length per routing table)	6
19	WR001_Clevis_Base	Wrist base clevis / distal shaft adapter, pitch joint	1
20	WR-004_Pivot_Pin_RevA	Wrist pivot pin D1.2 x 8.0 (pitch and yaw joints)	2
21	WR-002_Pitch_Link	Pitch link: integral pitch drum R2.2, distal yaw clevis	1
22	WR-005_Idler_Pulley_DoubleGroove	Wrist idler pulley, double groove, OD3.2 x 1.6, bore 1.2	4
23	WR003_Jaw_Carrier_Yaw_Link	Jaw carrier / yaw link: integral yaw drum R2.2, slider channel, jaw clevis	1
24	JA-003_Jaw_Pivot_Pin_RevB	Jaw pivot pin D1.0 h6 x 7.6, laser-welded/peened ends	1
25	JA-001_Jaw_Serrated_RevB	Grasper jaw, serrated, 15 long, angled cam slot (qty 2: upper + lower rotated 180 deg)	2
26	JA-004_Jaw_Slider_Yoke_RevB	Jaw actuation slider yoke: rear cable plate + twin arms, cross-pin bore	1
27	JA-005_Cam_Cross_Pin_RevB	Cam cross pin D0.5 x 3.2, press fit in slider, slip in jaw slots	1
28	JA006_Jaw_Pin_Reversing_Sleeve	Grooved PEEK sleeve on jaw pin: tang spacer + open-cable reversing pulley (qty 2)	2

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING | REVISION A

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TITLE:

MATERIAL:

DWG NO. LG-000_Laparoscopic_Grasper A2

WEIGHT:

SCALE:1:2 | SHEET 1 OF 1

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